

Nirma University

Institute of Technology

Semester End Examination (IR/RPR), January - 2025

Master of Computer Applications, Semester – I

6CS103CC22 Operating System

Roll/
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

- Instructions:
1. All questions are compulsory. (No Optional Questions)
 2. Use a section-wise separate answer book.
 3. Figure at right indicates full marks.
 4. Draw neat sketches wherever necessary.
 5. Assume suitable data if necessary.

SECTION-I

- Q-1 (A)** What is an operating system? Explain the types of operating systems. [08]
CLO1
BL2
- Q-1 (B)** Describe the role of different types of schedulers in the process execution with an appropriate diagram. [08]
CLO1
BL2
- Q-2 (A)** What is meant by process suspension? Explain briefly the various reasons for process suspension. [06]
CLO2
BL2
- Q-2 (B)** Differentiate FCFS (First come, first serve) and Round-robin uniprocessor scheduling algorithms based on the following characteristics. [06]
CLO2
BL3
- (i) Selection Function
 - (ii) Decision Mode
 - (iii) Throughput
 - (iv) Response time
 - (v) Effect on Processes
 - (vi) Possibility of Starvation
- Q-2 (C)** What is cache memory? With suitable diagrams, explain the cache structure and describe the cache read operation. [06]
CLO2
BL3
- Q-3 (A)** Differentiate consumable resources from reusable resources. [08]
CLO2
BL4
- Consider a system with 2 processes, P1 and P2. There are 4 units of resource R1 and 3 units of resource R2. Process P1 has requested 2 units of R1 and is holding 1 unit of R2. Process P2 has requested 1 unit of R2 and is holding 2 units of R1.
- Draw the necessary resource allocation graph to depict the scenario. Based on the resource allocation graph, find out if the system is in a deadlock or not. If it is in a deadlock, describe the method used to recover from the deadlock.

- Q-3 (B)** Consider the following page reference string: [08]
CLO2
BL4
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
How many page faults would occur for the following page replacement algorithm, considering 3 frames and 4 frames.
i) Optimal
ii) LRU

SECTION-II

- Q-4 (A)** Design a solution for producer-consumer problem using Semaphore [08]
CLO3
BL4
for finite buffer. (Write pseudo-code). Discuss your solution in detail.
- Q-4 (B)** What is Paging? Explain the paging mechanism in the Memory [08]
CLO3
BL3
Management Unit with an example.
- Q-5 (A)** Discuss the advantages and disadvantages of User-level threads (ULT) [06]
CLO3
BL3
with Kernel-level threads (KLT).
- Q-5 (B)** What is Interrupt? Why is it required in operating systems? Explain [06]
CLO3
BL3
various types of Interrupts.
- Q-5 (C)** Describe the process of taking action by the operating system when a [06]
CLO3
BL3
page fault occurs.
- Q-6 (A)** Suppose the Disk drive has 300 cylinders. The current position of the [08]
CLO4
BL4
head is 90. The queue of pending requests is
36,79,15,120,199,270,89,170
Calculate head movement for the following algorithms.
1. FCFS 2. SSTF
- Q-6 (B)** Consider a hypothetical 32-bit microprocessor having 32-bit [08]
CLO4
BL4
instructions composed of two fields. The first byte contains the opcode, and the remainder is an immediate operand or an operand address.
- **a.** What is the maximum directly addressable memory capacity (in bytes)?
 - **b.** Discuss the impact on the system speed if the Microprocessor bus has
 - 1. a 32-bit local address bus and a 16-bit local data bus,
 - 2. a 16-bit local address bus and a 16-bit local data bus.
 - **c.** How many bits are needed for the program counter and the instruction register?